

# Universal Linear Algebra Compliance Audit (Mode: NORMAL) (ALL)

**Audit Context:**

Mode: NORMAL

Reference: Episteme (Standard)

Matrix Size: 12

| Provider           | Operation                                                              | Domain                    | Result Summary                      |
|--------------------|------------------------------------------------------------------------|---------------------------|-------------------------------------|
| Generic JVM        | Apache Commons Math (Optimized)                                        | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM        | Colt (Optimized)                                                       | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM        | EJML (Optimized)                                                       | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (CARMA)        | Episteme (CARMA)                                                       | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (Standard)     | Episteme (Standard)                                                    | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (Strassen)     | Episteme (Strassen)                                                    | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (Standard JVM) | Episteme CPU (Dense)                                                   | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM        | Episteme CPU (Sparse)                                                  | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM        | Episteme CPU Foundation                                                | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM        | Episteme Distributed Linear Algebra Provider (LocalDistributedContext) | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM        | Episteme gRPC Remote (localhost:50051)                                 | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |

|                     |                                           |                           |                                     |
|---------------------|-------------------------------------------|---------------------------|-------------------------------------|
| Generic JVM         | JBlas (Optimized)                         | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (Native ND4J)   | ND4J (Native-Dense)                       | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (Native/Panama) | Native CPU-BLAS (Complex)                 | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (Native/Panama) | Native CPU-BLAS (Real)                    | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (FFM-BLAS)      | Native FFM-BLAS (Complex)                 | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (FFM-BLAS)      | Native FFM-BLAS (Real)                    | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| CPU (Panama/MPFR)   | Native MPFR Dense Linear Algebra Backend  | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM         | Native MPFR Sparse Linear Algebra Backend | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM         | Native SIMD (Complex)                     | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |
| Generic JVM         | Native SIMD (Real)                        | Linear Algebra Compliance | FAST/NORMAL/EXACT results collected |

# Operation Comparison: Apache Commons Math (Optimized) (ALL)

*Episteme Multimodal Audit | Apache Commons Math (Optimized)*

# Operation Comparison: Colt (Optimized) (ALL)

*Episteme Multimodal Audit | Colt (Optimized)*

## Operation Comparison: EJML (Optimized) (ALL)

*Episteme Multimodal Audit | EJML (Optimized)*

## Operation Comparison: Episteme (CARMA) (ALL)

*Episteme Multimodal Audit | Episteme (CARMA)*

**Operation Comparison: Episteme (Standard) (ALL)**

*Episteme Multimodal Audit | Episteme (Standard)*

## Operation Comparison: Episteme (Strassen) (ALL)

*Episteme Multimodal Audit | Episteme (Strassen)*



## Operation Comparison: Episteme CPU (Dense) (ALL)

*Episteme Multimodal Audit | Episteme CPU (Dense)*

## Operation Comparison: Episteme CPU (Sparse) (ALL)

*Episteme Multimodal Audit | Episteme CPU (Sparse)*

# Operation Comparison: Episteme CPU Foundation (ALL)

*Episteme Multimodal Audit | Episteme CPU Foundation*

# Operation Comparison: Episteme Distributed Linear Algebra Provider (LocalDistributedContext) (ALL)

*Episteme Multimodal Audit | Episteme Distributed Linear Algebra Provider (LocalDistributedContext)*

## Operation Comparison: Episteme gRPC Remote (localhost:50051) (ALL)

*Episteme Multimodal Audit | Episteme gRPC Remote (localhost:50051)*

## Operation Comparison: JBlas (Optimized) (ALL)

*Episteme Multimodal Audit | JBlas (Optimized)*

## Operation Comparison: ND4J (Native-Dense) (ALL)

*Episteme Multimodal Audit | ND4J (Native-Dense)*

## Operation Comparison: Native CPU-BLAS (Complex) (ALL)

*Episteme Multimodal Audit | Native CPU-BLAS (Complex)*



## Operation Comparison: Native CPU-BLAS (Real) (ALL)

*Episteme Multimodal Audit | Native CPU-BLAS (Real)*

## Operation Comparison: Native FFM-BLAS (Complex) (ALL)

*Episteme Multimodal Audit | Native FFM-BLAS (Complex)*

## Operation Comparison: Native FFM-BLAS (Real) (ALL)

*Episteme Multimodal Audit | Native FFM-BLAS (Real)*

# Operation Comparison: Native MPFR Dense Linear Algebra Backend (ALL)

*Episteme Multimodal Audit | Native MPFR Dense Linear Algebra Backend*

# Operation Comparison: Native MPFR Sparse Linear Algebra Backend (ALL)

*Episteme Multimodal Audit | Native MPFR Sparse Linear Algebra Backend*

## Operation Comparison: Native SIMD (Complex) (ALL)

*Episteme Multimodal Audit | Native SIMD (Complex)*

## Operation Comparison: Native SIMD (Real) (ALL)

*Episteme Multimodal Audit | Native SIMD (Real)*